

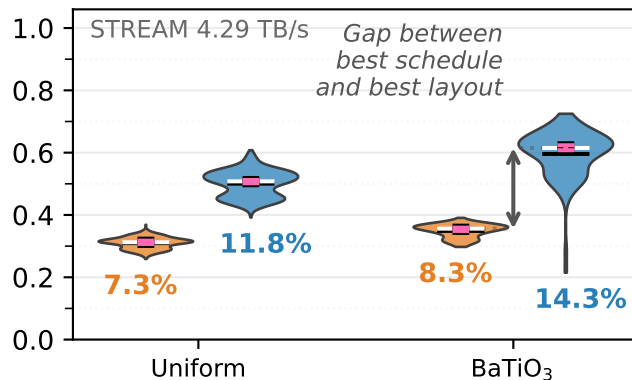
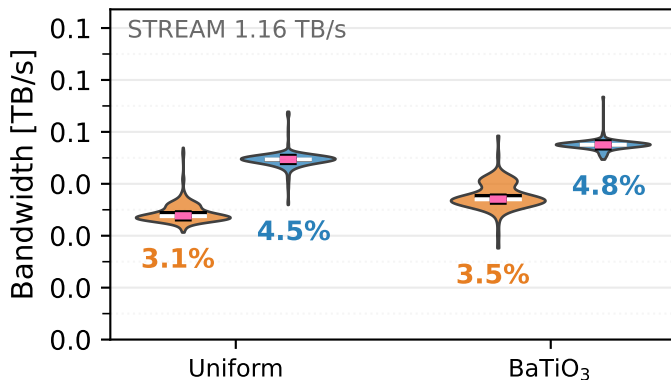
Gather-Accumulate-Scatter

% annotations relative to STREAM peak bandwidth

Original: $y[\sigma(i)] += x[i]$; Shuffled: $y[\sigma(\pi(i))] += x[\pi(i)]$, $\pi = \text{argsort}(\sigma)$

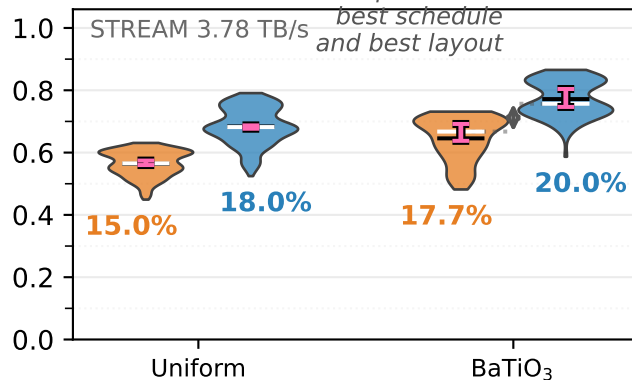
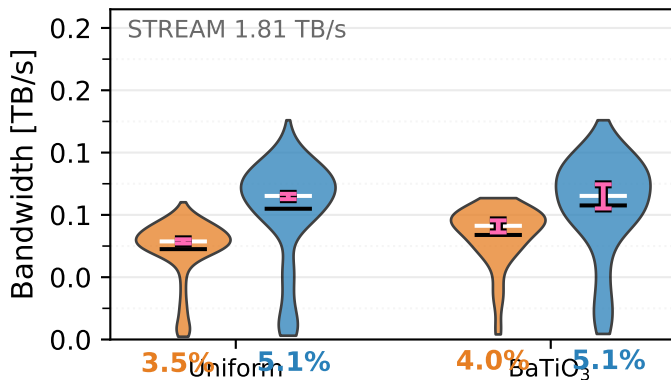
MI300A Zen CPU

MI300A GPU



GH200 Grace CPU

GH200 Hopper GPU



Original Order Sorted (Shuffle)